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Total Domination in Geometric Unit Disk Graphs

Sangram K. Jena*

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Abstract

Let $G = (V, E)$ be an undirected graph. We call $D_t \subseteq V$ a total dominating set (TDS) of G if each vertex $v \in V$ has a dominator in D_t other than itself. Here we consider the TDS problem in geometric unit disk graphs, where the objective is to find a minimum cardinality total dominating set for an input graph. We prove that the problem of deciding whether a geometric unit disk graph (UDG) G has a TDS of cardinality at most k (here k is a positive integer) is NP-complete. Next, we propose an almost linear time 8-factor approximation algorithm in geometric UDGs for the TDS problem. It is an improvement over the best-known 10-factor approximation algorithm with the same running time available in the literature [M. V. Marathe, H. Breu, H. B. Hunt III, S. S. Ravi and D. J. Rosenkrantz, *Simple heuristics for unit disk graphs*, *Networks*, 25(2):59–68, 1995]. We also show that the TDS problem admits a polynomial-time approximation scheme in geometric UDGs.

keywords: Total dominating set, approximation algorithm, PTAS, unit disk graph

1 Introduction

Let us consider a simple undirected graph $G = (V, E)$. The *open neighborhood* (resp. *closed neighborhood*) of a vertex $v \in V$ is the set $N_G(v) = \{u \in V : uv \in E\}$ (resp. $N_G[v] = N_G(v) \cup \{v\}$). A *dominating set* (DS) of G is a subset $D \subseteq V$ such that for each vertex $v \in V$, $|D \cap N_G[v]| \geq 1$. A *total dominating set* (TDS) is a subset $D_t \subseteq V$ of G such that for each vertex $v \in V$, $|D_t \cap N_G(v)| \geq 1$. Therefore, a vertex $v \in D$ (dominating set) dominates all its neighbors and itself whereas a vertex $v \in D_t$ (total dominating set) dominates all its neighbors other than itself. The objective of TDS (resp. DS) problem is to find a minimum size subset $D_t \subseteq V$ (resp. $D \subseteq V$) such that D_t (resp. D) dominates all the vertices in V .

The intersection graph of equal-radii disks with known position in the plane is called a *geometric unit disk graph* (UDG). Let $P = \{p_1, p_2, \dots, p_n\}$ be a set of n points in $\mathbb{R}^2$. Let $d_i = \{p \in \mathbb{R}^2 | d(p_i, p) \leq \frac{1}{2}\}$ be a disk of radius $\frac{1}{2}$ with centered at p_i for $1 \leq i \leq n$, where $d(.,.)$

denotes Euclidean distance between two points in $\mathbb{R}^2$. Let $S = \{d_1, d_2, \dots, d_n\}$ be the set of n equal-radii circular disks. We define the geometric UDG $G = (V, E)$ corresponding to the point set P as follows: each vertex $v_i \in V$ corresponds to the point $p_i \in P$, and $v_i v_j \in E$ if and only if the disks d_i and d_j intersect i.e., the Euclidean distance between p_i and p_j is at most one unit.

One of the applications of the problem is identifying the location of monitoring devices, such as surveillance cameras or fire alarms, to safeguard a system that can be modeled by total domination in graphs. The problem of placing monitoring devices in a system, where each monitoring device can be placed in such a way that every site in the system (including the monitors themselves) is adjacent to one of the monitoring devices. In this case, placing the monitoring devices in each solution point of the total domination of the system solves the problem.

1.1 Related Work

In 1980, Cockayne et al. [6] introduced the total domination problem and proved that for any connected graph G of $n(\geq 3)$ vertices the cardinality of minimum total dominating set, denoted by λ_t , is less than or equal to $\frac{2}{3}n$ i.e., $\lambda_t \leq \frac{2}{3}n$. Brigham et al. [2] proved that the total domination number is exactly $\frac{2}{3}n$ for the connected graph G of order $n(\geq 3)$, where G is either C_3 (cycle graph of 3 vertices), C_6 or 2-corona of some connected graph. Sun [21] proved the bound to $\lambda_t \leq \lfloor \frac{4}{7}(n+1) \rfloor$, for connected graphs having order n with minimum degree at least 2. Chvátal and McDiarmid [5] and Tuza [23] independently proved a theorem concerning transversals in hypergraphs, which gives a bound on total domination number. The bound is $\lambda_t \leq \frac{n}{2}$ for the graphs with order n and minimum degree at least 3. For the graphs with minimum degree at least 4, Thomassé and Yeo [22] proposed a result for hypergraphs, which bounds the total domination number by $\lambda_t \leq \frac{3}{7}n$. In [8], DeLaViña et al. proved that the total domination number of any connected graph is equal to the total domination number of a spanning tree of the same graph. Another interesting aspect of trees concerning total domination is that it is possible to characterize some vertices that are in every minimum total dominating set or not in any total dominating set [7]. Furthermore, Haynes and Henning established three equivalent conditions for a tree to have a unique minimum total dominating set [10]. Chellali and Haynes [4] proved $\lambda_t \geq \frac{n+2-\ell}{2}$ for a

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nontrivial tree of n vertices with ℓ leaves. Dorfling et al. [9] bound the total domination number of planar graphs having different diameters and radius. Pfaff et al. [20] showed that computing λ_t for general graphs is NP-complete. In the same paper, they also showed that calculating λ_t for bipartite graphs remains NP-complete. However, a linear-time algorithm exists for computing λ_t in tree graph [19]. The total domination number in the case of star graphs, complete graphs, binary star graphs, and complete bipartite graphs is 2 [1]. In the same article, they have observed that for cycles and paths, the total dominating set can be found in polynomial time. They have also established a set of relations between (i) λ_t and the maximum degree, and (ii) λ_t and the cut vertices of the graph. See [1, 11, 12, 13, 14] for a detailed survey on the TDS problem.

Dominating set and its many variants such as weighted dominating set, connected dominating set, etc. have been extensively studied in the literature due to their wide range of applications in wireless networks. A wireless network can be modeled as a unit disk graph. Though the (variant) domination problems are still NP-hard for unit disk graphs, unlike in general graphs the problems admit constant approximation algorithms and approximation schemes [25, 26, 29, 30, 31, 24, 27, 28].

1.2 Our Contribution

In this paper, we consider the total dominating set problem in geometric UDGs. In Section 2, we prove that the decision version of the TDS problem is NP-complete in geometric UDGs. We propose an almost linear time 8-factor approximation algorithm for the same problem in Section 3. We also show that the problem admits a PTAS in Section 4. Finally, we conclude the paper in Section 5.

2 NP-Completeness

In this section, we prove that the TDS problem in UDGs belongs to the class NP-complete. In [18], Lichtenstein proved that the planar vertex cover problem is NP-hard by showing a polynomial-time reduction from the planar 3SAT problem to the planar vertex cover problem. In the reduction from an arbitrary instance of the planar 3SAT problem to an instance of planar vertex cover problem, the degree of the graph is at most 3. Therefore the planar vertex cover problem is NP-hard for the graphs with the degree at most 3. We prove the NP-hardness result of the TDS problem in UDGs by showing a polynomial-time reduction from the vertex cover problem in planar graphs of degree at most 3 to it. Now, we define the decision version of the TDS problem in UDGs and vertex cover problem in planar graphs of degree at most 3 as follows:

The TDS problem in UDGs (TDS-UDG)

Given a unit disk graph G and an integer $k(> 0)$, does there exist a TDS of size at most k ?

The VC problem in planar graphs (VC-PLA)

Given a planar graph G with degree at most 3 and an integer $k(> 0)$, does G has a VC of size at most k ?

Lemma 1 ([32]) *Let $G = (V, E)$ be a planar graph with maximum degree 3. The graph G can be embedded in linear time on a planar grid of size 4×4 using $O(|V|^2)$ area such that the coordinate of each vertex $v \in V$ is $(4i, 4j)$ for some integers i, j and each edge $e \in E$ is a finite sequence of consecutive axis-parallel line segments of length 4 units along the grid lines.*

Lemma 2 *For a given VC-PLA instance $G = (V, E)$ with at least one edge, an instance $G' = (V', E')$ of TDS-UDG can be constructed in polynomial-time.*

Proof. Let $V = \{v_1, v_2, \dots, v_n\}$ and $E = \{e_1, e_2, \dots, e_m\}$. The construction of G' from the graph G is described in four steps as follows.

Step 1: (Embedding) We first embed G into a planar grid of size $4n \times 4n$ using Lemma 1. On the embedding, each of the vertex $v_i \in V$ becomes grid point p_i and each edge $e_j \in E$ become a finite sequence of connected axis-parallel line segment(s) of length four units along the grid lines. Assume that ℓ is the total number of line segments used in the embedding. We call the point p_i corresponding to the vertex $v_i \in V$ ($i = 1, 2, \dots, n$) in the embedding as *vertex points* (see Figure 1(a) and 1(b)). Let N be the set of vertex points. Therefore, $N = \{p_i \mid v_i \in V\}$ and $|N| = |V| (= n)$.

Step 2: (Extra points) In this step, we add some extra points on each of the ℓ line segments (obtained in embedding step) so that unit disks centered on these points and grid points (see embedding step) form a unit disk graph as follows: **(a)** for each edge $p_i p_j$ with only one line segment i.e., length of the edge is 4 units, we add five points at distance 0.98, 1.49, 2, 2.51, 3.02 units from p_i (see edge $p_4 p_6$ in Figure 1(c)), and **(b)** for each edge $p_i p_j$ with more than one segment i.e., length of the edge is greater than 4 units, (i) add a point on each of the grid point on the edge other than the vertex point and name it as grid point (see square points in Figure 1(c)), and (ii) we add four points on each of the line segments connected with p_i and p_j at distances 1, 1.75, 2.5, 3.25 units from p_i and p_j , and for other line segments we add three points at distance 1 units from each other excluding the *grid*

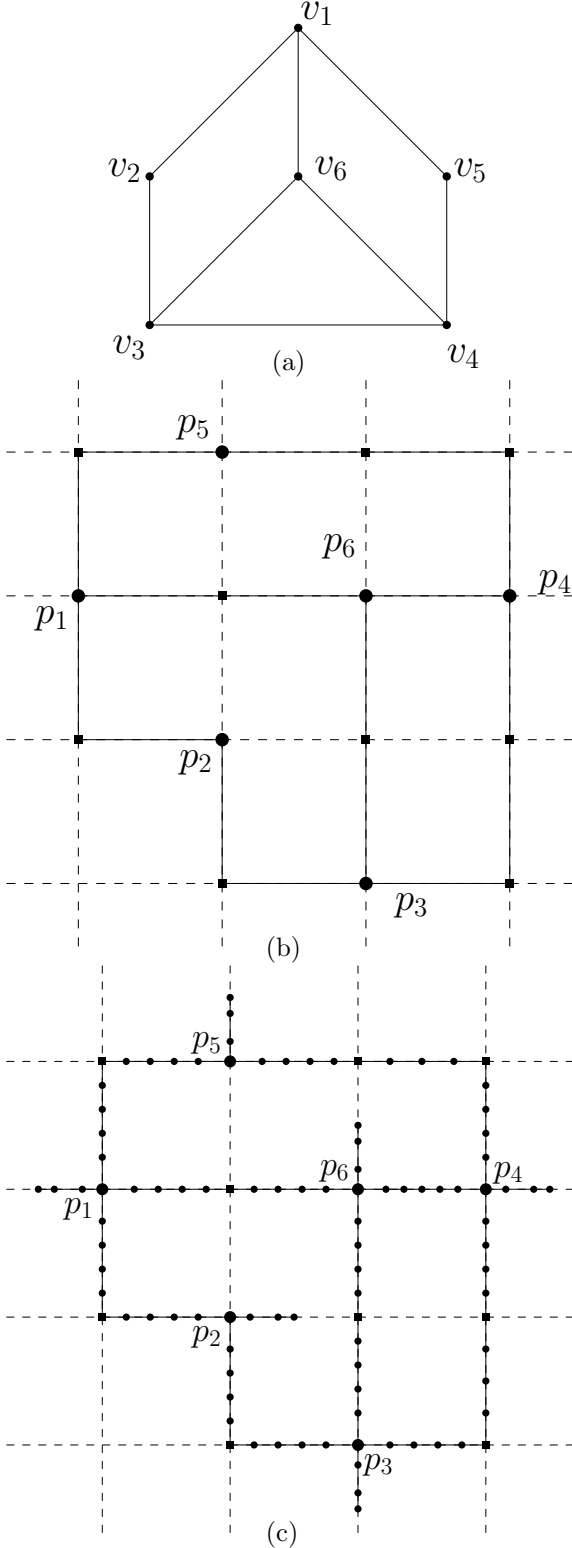


Figure 1: (a) A planar graph G with maximum degree 3, (b) its embedding on a 4×4 grid, and (c) construction of a UDG from the embedding.

points (see the edge p_3p_4 in Figure 1(c)). Let A be the set of all points added in this step. Therefore, $|A| = 4\ell + m$ (here ℓ is the total number of line segments in the embedding and m is the number of edges in G).

Step 3: (Support point) Add a new line segment of length 1.4 units at each of the vertex points p_i without coinciding with the line segments that had already been drawn in the embedding. Observe that the addition of such a line segment is possible without losing the planarity as the maximum degree of G is 3. We add three points x_i, y_i, z_i on each of these line segments at distances 0.3, 1.1, and 1.4 units from the corresponding vertex point p_i . Let S be the set of all points added in this step. Therefore, $|S| = 3n$ (here n is the number of vertices in G).

Step 4: (Construction of UDG) We construct a UDG $G' = (V', E')$, where $V' = N \cup A \cup S$ and $E' = \{u'v' : u', v' \in V' \text{ and the Euclidean distance between } u' \text{ and } v' \text{ is at most 1 unit}\}$ (see Figure 1(c)).

From Lemma 1, $\ell = O(n^2)$. Therefore both $|V'|$ and $|E'|$ are bounded by $O(n^2)$. Hence, G' can be constructed in polynomial-time. $\square$

Theorem 3 TDS-UDG belongs to the class NP-complete.

Proof. Let $T \subseteq V$ be an arbitrary subset of vertices and $k(> 0)$ be an integer. Observe that, we can verify whether T is a total dominating set such that $|T| \leq k$ or not in polynomial-time. Therefore, TDS-UDG $\in$ NP.

To prove NP-hardness of TDS-UDG, we will use polynomial time reduction of VC-PLA to it. We construct an instance $G' = (V', E')$ of TDS-UDG from an arbitrary instance $G = (V, E)$ of VC-PLA in polynomial time using the steps mentioned in Lemma 2. Next, we prove the following claim to complete the proof of NP-hardness of TDS-UDG.

Claim: G has a vertex cover C with $|C| \leq k$ if and only if G' has a total dominating set T with $|T| \leq k + 2\ell + 2n$.

Necessity: Let $C \subseteq V$ be a vertex cover of G such that $|C| \leq k$. Let $N' = \{p_i \in N \mid v_i \in C\}$, i.e., N' is the set of vertices (or vertex points) in G' that correspond to the vertices in C . From each segment, we choose 2 vertices (extra points) from A and corresponding to each vertex point, we choose 2 points (support points) from S , in the embedding. The set of chosen vertices, say $A'(\subseteq A)$, $S'(\subseteq S)$, together with N' will form a TDS of the desired cardinality in G' . We now discuss the process of obtaining the set A' . Initially $A' = \emptyset$. As C is a vertex cover, every edge in G has at least one of its

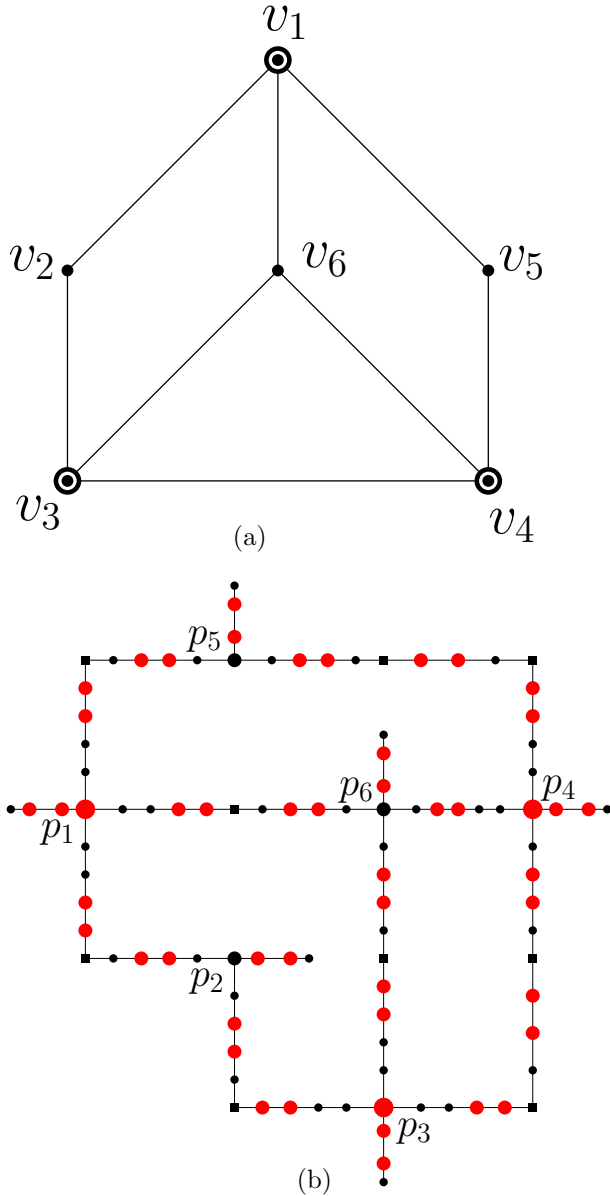


Figure 2: (a) A vertex cover $\{v_1, v_3, v_4\}$ in G , and (b) the construction of A' in G' (the tie between v_3 and v_4 is broken by choosing v_3)

end vertices in C . Let $v_i v_j$ be an edge in G and $v_i \in C$ (choose any of them arbitrarily if both v_i and v_j are in C). Note that the edge $v_i v_j$ is represented as a sequence of line segments in the embedding. Start traversing the segments (of $v_i v_j$) from p_i , where p_i corresponds to v_i , and add two consecutive vertices by leaving two consecutive vertices in between starting from p_i to A' in the traversal (see $p_4 p_5$ in Figure 2(b)). The red bold vertices are part of A' while traversing from p_4 .

Apply the above process to each edge in G . Observe that the cardinality of A' is 2ℓ as we have chosen 2 vertices from each segment in the embedding. Next, we

choose $2n$ points from S in $S' = \{x_i, y_i : p_i \in N\}$. Let $T = N' \cup A' \cup S'$. Now, we argue that T is a total dominating set in G' .

For each point $p_i \in N$, p_i is dominated by x_i , x_i is dominated by y_i , y_i is dominated by x_i and z_i is dominated by y_i . So, the sets N and S satisfies total domination condition. Now it is remaining to prove that the set A satisfies total domination condition. Observe the way we have chosen points from A in T , with a gap of two consecutive points, two consecutive points are chosen in T . For each point $p_i \in T$, p_i dominates $N_G(p_i) \in A$ and the selected points of A in T can total dominate all the remaining points of A (see the edge $p_1 p_2$ in Figure 2(b)).

Therefore, T is a TDS in G' and $|T| = |N'| + |A'| + |S'| \leq k + 2\ell + 2n$.

Sufficiency: Let $T \subseteq V'$ be a TDS of size at most $k + 2\ell + 2n$. We prove that G has a vertex cover of size at most k with the help of the following claims.

- (i) Out of three support points associated with each $p_i \in N$, at least two points belongs to T , i.e., $|S \cap T| \geq 2n$.
- (ii) Every segment in the embedding must contribute at least two points to T and hence $|A \cap T| \geq 2\ell$, where ℓ is the total number of segments in the embedding.
- (iii) If p_i and p_j correspond to end vertices of an edge $v_i v_j$ in G , and if both p_i, p_j are not in T , then there must be at least $2\ell' + 1$ vertices in T from the segment(s) representing the edge $v_i v_j$, where ℓ' is the number of segments representing the edge $v_i v_j$ in the embedding.

Claim (i) directly follows from the definition of total dominating set. Observe that we added points x_i, y_i, z_i such that p_i is adjacent to x_i , x_i is adjacent to y_i , and y_i is adjacent to z_i in G' , i.e., $\{p_i x_i, x_i y_i, y_i z_i\} \subseteq E'$ for each i . Hence, y_i must be in T as y_i is the only vertex which can dominate z_i and either x_i or z_i must be in T to dominate y_i . Therefore, any total dominating set of G' must contain two support points in T out of three support points associated with $p_i, 1 \leq i \leq n$, i.e., $|S \cap T| \geq 2n$.

Claim (ii) follows from the fact that only consecutive points are adjacent (in G') on any segment in the embedding. Let η be a segment in the embedding having vertices q_i, q_{i+1}, q_{i+2} , and q_{i+3} . On contrary, assume that η has only one of its vertices in T . Note that only q_i can not be in T . If q_i present in T , then q_{i+2} is not dominated by any point, which is a contradiction to the fact that T is a TDS. If q_{i+1} is the only point in T then q_{i+1} is not dominated by any point. If q_{i+2} will be chosen as the only point from η in T then q_{i+2} is not dominated by any other point and finally if q_{i+3} will be

chosen then q_{i+1} is not dominated by any other point. In all cases, we arrived at a contradiction.

Claim (iii) follows from the definition of the total dominating set that any point chosen in the solution set dominates all its neighbors other than itself. Here any point selected from a segment in T has exactly two neighbors other than itself. So, it can dominate at most 2 points. There are ℓ' segments between two node points p_i and p_j having $4\ell' + 1$ number of points and both p_i and p_j are not in T . So, the minimum number of points required in T to ensure total domination is $\left\lceil \frac{4\ell'+1}{2} \right\rceil = 2\ell' + 1$.

Now, we will show that, by removing and/or replacing some vertices in T , a set of at most k points from N can be chosen such that the corresponding vertices form a vertex cover in G . The vertices in S account for $2n$ vertices in T (due to Claim (i)). Let $T = T \setminus S$ and $C = \{v_i \in V \mid p_i \in T \cap N\}$. If any edge $v_i v_j$ in G has none of its end vertices in C , then we do the following: consider the sequence of segments representing the edge $v_i v_j$ in the embedding. Since both p_i and p_j are not in T , there must exist a segment having three vertices in T (due to Claim (iii)). Consider the segment having its three vertices in T . Delete any one of the vertices on the segment and introduce p_i (or p_j). Update C and repeat the process till every edge has at least one of its end vertices in C . Due to Claim (ii), C is a vertex cover in G with $|C| \leq k$. Therefore, TDS-UDG $\in$ NP-hard. As TDS-UDG $\in$ NP and TDS-UDG $\in$ NP-hard, TDS-UDG $\in$ NP-complete. $\square$

3 Approximation Algorithm

In this section, we propose an 8-factor approximation algorithm for the TDS problem in UDGs. The worst-case time complexity of our proposed algorithm is $O(n \log k)$, where k is the size of the output of our algorithm. Note that, if there exists an r -factor approximation algorithm for the dominating set problem then a $2r$ -factor approximation algorithm for the total dominating set problem can be obtained by adding one neighbor for each vertex in the dominating set. A 10-factor approximation algorithm is available in the literature for the TDS problem in UDGs [26]. In this section, we improved the approximation factor of the TDS problem in UDGs to 8. The running time of the 8-factor approximation algorithm is $O(n \log k)$.

Lemma 4 [26] *Let C be a unit disk centered at a point p . If S is a set of independent¹ unit disks such that every unit disk in S contains the point p then $|S| \leq 5$.*

Lemma 5 *Let C' and C'' be two unit disks centered at points p and q respectively, such that $d(p, q) = 1$. If S*

¹Two disks are independent if the distance between their centers is greater than one.

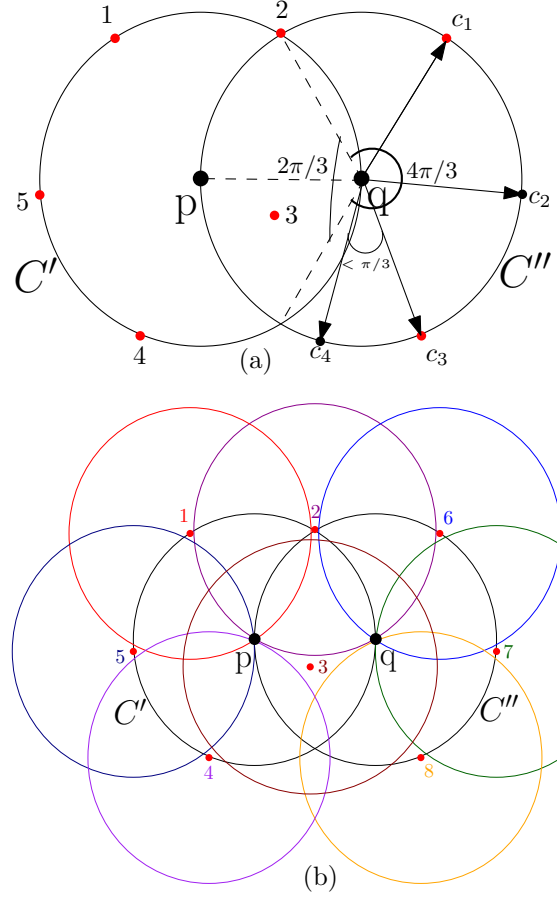


Figure 3: Illustration of Lemma 5.

is a set of independent unit disks containing the point p and/or q , then $|S| \leq 8$.

Proof. Let $S_p, S_q \subseteq S$ be the sets of independent unit disks containing the point p and q respectively. This implies $S = S_p \cup S_q$. Note that, the cardinality of S will increase if the distance between p and q increases. Therefore, without loss of generality, we assume that the points p and q are one unit distance apart, i.e., the point q lies on the boundary of the unit disk C' centered at p (see Figure 3 for reference). Now, we consider the following two cases:

Case (i): $|S_p| \leq 3$. From Lemma 4, we observed that $|S_q| \leq 5$. Hence, $|S_p \cup S_q| \leq 8$.

Case (ii): $|S_p| > 3$. It is known that $|S_p| \leq 5$ (see Lemma 4). So $|S_p|$ is either 4 or 5.

(a) Assume that $|S_p| = 5$. Now it is remaining to prove that, there does not exist more than 3 independent unit disks in the set (say S'_q) $\subseteq S_q$, which contains the point q but does not contain the point p , i.e., $|S'_q| = |S_q| - |S_p \cap S_q| \leq 3$. For the sake of argument assume that $|S'_q| = 4$, that means, there exist 4 unit disks centered at c_i , $1 \leq i \leq 4$ in S'_q , which contains the point q . Let us denote the ray $\overrightarrow{qc_i}$ by r_i ,

($1 \leq i \leq 4$). Since there are 4 rays coming out from q and the disks centered at c_i , ($i = 1, 2, 3, 4$) does not contains p , there exist one pair of rays r_i and r_j such that the angle between them is at most $\pi/3$ (see Figure 3(a)), which leads to a contradiction that the disks centered at c_i , ($i = 1, 2, 3, 4$) are independent. Thus, $|S'_q| \leq 3$, which implies $|S_p \cup S_q| \leq 8$ (see Figure 3(b)).

- (b) Assume that $|S_p| = 4$. Now it is sufficient to prove that $|S'_q| < 5$, to prove $|S_p \cup S_q| \leq 8$. For the sake of argument assume that $|S'_q| = 5$, that means, there exist 5 independent unit disks in the set S'_q , which contains the point q but does not contain the point p . Following a similar argument as in case (a), it leads to a contradiction. Thus, $|S'_q| < 5$, which implies $|S_p \cup S_q| \leq 8$.

In either of the cases, we proved that $|S_p \cup S_q| \leq 8$. Therefore, $|S| \leq 8$. $\square$

Let P denote the set of n points (center of the disks) given in the plane $\mathbb{R}^2$. We use $\Delta(S)$ to denote the set of unit disks centered at the points in $S \subseteq P$. A dominating set in unit disk graphs (DS-UDG) $D \subseteq P$ of the set of disks $\Delta(P)$ is said to be an independent DS-UDG if for each pair $p, q \in D$, $p \notin N_G(q)$.

The procedure of generating a TDS-UDG for a given points set P in $\mathbb{R}^2$ is described in Algorithm 1.

Algorithm 1 Total dominating set in P

Require: A set of disks $\Delta(P)$

Ensure: A total dominating set T of $\Delta(P)$

```

1:  $D \leftarrow \emptyset$ , and  $T \leftarrow \emptyset$ 
2: while ( $P \neq \emptyset$ ) do
3:   choose an arbitrary point  $p \in P$ 
4:    $D \leftarrow D \cup \{p\}$ ;  $P \leftarrow P \setminus N_G[p]$ 
5: end while
6: for every  $p \in D$  do
7:   if  $N_G(p) \cap T = \emptyset$  then
8:     let  $q \in N_G(p)$ 
9:      $T = T \cup \{q\}$ 
10:  end if
11: end for
12:  $T = T \cup D$ 
13: return  $T$ 
```

Lemma 6 T returned by Algorithm 1 is a TDS-UDG for the set of unit disks $\Delta(P)$.

Proof. Before finding a TDS-UDG T , Algorithm 1 find an independent DS-UDG D (see **while** loop in line number 2 of the algorithm), which ensures domination for the set of unit disks $\Delta(P)$ and total domination for the points $P \setminus D$. Next, to obtain total domination in D , for each point $p \in D$ the algorithm ensures the existence of

a point $q \in N_G(p)$ in T (see **for** loop in line number 6 of the algorithm). Therefore the selected points in T along with D ensures total domination for the set of unit disks $\Delta(P)$. $\square$

Lemma 7 $|T| \leq 8|OPT|$, where OPT is a TDS-UDG for the unit disks $\Delta(P)$ of minimum size.

Proof. Consider an arbitrary point $p \in OPT$. As OPT is a TDS of minimum size there must exist a point $q \in OPT$ such that $p \in N_G(q)$ and the point q ensures the domination for the point p . Now consider both the unit disks centered at p and q . From Lemma 5, it is proved that there exist at most 8 unit disks in an independent DS-UDG D of $\Delta(P)$ that can contain the points p and/or q .

Now the cardinality of T follows from the fact that for every pair $p, q \in OPT$ such that $p \in N_G(q)$, there may exist at most 8 unit disks in an independent DS-UDG D of $\Delta(P)$ that can contain the points p and/or q . Our algorithm chooses at most one neighbor for each point in D in to the solution set T , to make T as a TDS-UDG (see line numbers 6-12 of Algorithm 1). So for 8 points in D at most 16 points may be chosen in T with respect to two points $p, q \in OPT$. which leads at most 16 points are chosen by Algorithm 1 against 2 points (namely, p and q) chosen in the optimal solution of the TDS, i.e., $|T| \leq \frac{16}{2}|OPT|$. Therefore, $|T| \leq 8|OPT|$. $\square$

Lemma 8 The worst-case time required to generate a TDS-UDG for the set of disks $\Delta(P)$ by Algorithm 1 is $O(n \log k)$, where k is the size of the output.

Proof. We now describe the time complexity of Algorithm 1 for computing a TDS-UDG T of $\Delta(P)$ as follows. Let us assume that $\mathcal{R}$ is an axis parallel rectangular region containing the points in P . We partition $\mathcal{R}$ into grid cells of size 1×1 . A point $p_i = (x_i, y_i) \in P$ lies in the grid cell indexed by $[\lfloor x_i \rfloor, \lfloor y_i \rfloor]$ for $i = 1, 2, \dots, n$. Each grid cell is attached with a list of points in P that belong to that cell. We construct an independent dominating set D for UDG corresponding to $\Delta(P)$. While considering a point $p_i \in P$, we inspect all members of D which are attached to all 9 cells $[\alpha, \beta]$, where $\lfloor x_i \rfloor - 1 \leq \alpha \leq \lfloor x_i \rfloor + 1$ and $\lfloor y_i \rfloor - 1 \leq \beta \leq \lfloor y_i \rfloor + 1$. If there does not exist any unit disk d in D that contains the point p_i , we add p_i in D . Observe that, at the end of considering all the points in P , D will be an independent DS-UDG for the set of disks in $\Delta(P)$. Initially, take $T = \emptyset$. Now, for each point $p \in D$, if there does not exist any point q in T such that $q \in N_G(p)$, then add q in T , and the existence of q is guaranteed, otherwise finding TDS for this given instance is impossible as the point p is an isolated point. Finally, update $T = T \cup D$. After ensuring the existence of a point $q \in N_G(p)$ for each point $p \in D$, observe that T is a TDS-UDG for the

set of disks in $\Delta(P)$. Note that, (i) a grid cell may contain at most 6 points in T , and (ii) the number of grid cells to be inspected while processing a point $p_i \in P$ is at most 9. We use a height-balanced binary tree to store the indices of the grid cells containing a non-zero number of points in T . Thus, the time complexity for processing a point $p \in P$ is $O(\log k)$, where $k = |T|$ and $|P| = n$. $\square$

Theorem 9 *Algorithm 1 is an 8-factor approximation algorithm for the TDS-UDG problem. The running time of the algorithm is $O(n \log k)$, where n is the input size and k is the output size.*

Proof. Follows from Lemma 6, Lemma 7, and Lemma 8. $\square$

4 Approximation Scheme

In this section, we propose a polynomial-time approximation scheme (PTAS) for the TDS problem in geometric UDGs. We use the shifting strategy [15] technique to propose a PTAS. Let P be a point set (centers of the disks) given in a rectangular region $\mathcal{R}$ along with a fixed integer $k \geq 1$.

We use a two-level nested shifting strategy to propose a PTAS for the said problem. The first level of shifting strategy is applied in the vertical direction on $\mathcal{R}$. There are k iterations in the first level and the i -th iteration ($1 \leq i \leq k$) partition the region $\mathcal{R}$ into many vertical strips, where the first strip is of width $2i$, and remaining strips other than the last strip are of width $2k$. The width of the last strip may be less than $2k$. Without loss of generality, assume that each point lying on the left boundary of a strip belongs to its left adjacent strip (i.e., every strip is left open and right closed). Now consider all the non-empty vertical strip $\mathcal{V}$, and apply the second level of shifting strategy in the horizontal direction. In the second level of shifting strategy, the j -th iteration ($1 \leq j \leq k$) partition each non-empty vertical strip $\mathcal{V}$ into square/rectangular cells of size $2j \times \ell$ for the first cell and $2k \times \ell$ for all other cells, where ℓ defines the width of the strip $\mathcal{V}$ ($\ell = 2i$ for the first strip and $\ell = 2k$ for all other strips except for the last strip).

We consider each non-empty $2k \times 2k$ square (conceptually extending the smaller cells into $2k \times 2k$ square) and find the optimal solution of each square. The union of the optimal solution of each $2k \times 2k$ square gives a feasible solution of each strip $\mathcal{V}$. Finally, we take the union of the solution of each non-empty vertical strip to get a feasible solution of the problem in a single iteration, i.e., (i, j) -th iteration. In the same process, we get the feasible solutions of all the iterations in the first level. We report the solution T , having minimum cardinality among all the solutions generated in each iteration as the solution of the TDS-UDG problem.

Now, we discuss the procedure of getting an optimal solution from each $2k \times 2k$ square. We first partition the cell of size $2k \times 2k$ into $(\lceil 2\sqrt{2}k \rceil)^2$ sub-cells. The size of each sub-cell is $\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$. Observe that, choosing any two points inside a sub-cell of size $\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$ ensures total domination for all unit disks centered in that sub-cell. Hence, the maximum number of points required to ensure total domination in a square of size $2k \times 2k$ is $2(\lceil 2\sqrt{2}k \rceil)^2$. Therefore, we have to check all possible combinations of points up to $2(\lceil 2\sqrt{2}k \rceil)^2$ to get an optimal solution in a cell χ of size $2k \times 2k$. Note that, along with the points inside a cell χ , the points within 1 unit apart from χ also play a crucial role to get an optimum solution of χ . Let n_χ be the number of points in P whose corresponding disks have a portion in the cell χ (n_χ includes the points inside χ along with the points within 1 unit apart from χ). Then, we have to choose at most $O(n_\chi^{2(\lceil 2\sqrt{2}k \rceil)^2})$ combinations of points for getting the optimum solution for the TDS-UDG problem in a cell χ of size $2k \times 2k$. Since the points in P centered in a cell are disjoint from that of the other cells, and a point in P can participate in computing the optimum solution of at most 9 cells, we have the following result.

Lemma 10 *The total time required for the (i, j) -th iteration of the algorithm is $O(n^{2(\lceil 2\sqrt{2}k \rceil)^2})$.*

Proof. The feasible solution of the (i, j) -th iteration is the union of the optimum solutions of all the cells constructed in that iteration. Finally, the algorithm returns the minimum among the k^2 feasible solutions corresponding to k^2 iterations. $\square$

Theorem 11 *Given a set P of n points (center of the unit disks) in $\mathcal{R}$ and an integer $k \geq 1$, a total dominating set of size at most $(1 + \frac{1}{k})^2 \times |OPT|$ can be computed in $O(k^2 n^{2(\lceil 2\sqrt{2}k \rceil)^2})$ time, where OPT is the optimum solution.*

Proof. Using the shifting strategy analysis given by Hochbaum and Maass [15], we analyze the approximation factor of our algorithm. Let OPT be an optimum solution for the TDS-UDG problem for the point set P , and $OPT' \subseteq OPT$ be such points chosen in OPT , which totally dominate some points outside the boundary of all the cells in an (i, j) -th iteration. Let T be a solution obtained by our algorithm in an iteration. Then, $|T| \leq |OPT| + |OPT'|$. For all the iterations of (i, j) ($1 \leq i, j \leq k$), we have $\sum_{i=1}^k \sum_{j=1}^k |T| \leq k^2 |OPT| + \sum_{i=1}^k \sum_{j=1}^k |OPT'|$. Since any point from a cell χ chosen in OPT can dominate points from no more than one horizontal strip (or vertical strip), and at most k times each horizontal (or vertical) boundary appears throughout the algorithm,

we have $\sum_{i=1}^k \sum_{j=1}^k |OPT'| \leq k|OPT| + k|OPT|$. Thus,

$$\sum_{i=1}^k \sum_{j=1}^k |T| \leq k^2|OPT| + 2k|OPT| = (k^2 + 2k)|OPT|.$$

Therefore, $\min_{i=1}^k \sum_{j=1}^k |T| \leq (1 + \frac{1}{k})^2 \times |OPT|$. The time complexity result follows from Lemma 10. $\square$

5 Conclusion

In this article, we have considered the minimum total dominating set (TDS) problem in geometric unit disk graphs. We showed that the TDS problem belongs to the class NP-complete. We proposed an almost linear time 8-factor approximation algorithm, which is an improvement over the best-known 10-factor approximation algorithm with the same running time available in the literature [26]. We also proved that the TDS problem in geometric UDGs admits a PTAS.

References

- [1] D. Amos and E. DeLaVina. On Total Domination in Graphs. *Senior Project, University of Houston Downtown*, 2012.
- [2] R. C. Brigham, J. R. Carrington, and R. P. Vitray. Connected Graphs with Maximum Total Domination Number. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 34:81–96, 2000.
- [3] Paz Carmi, Gautam K. Das, Ramesh K. Jallu, Subhas C. Nandy, Prajwal R. Prasad, and Yael Stein. Minimum Dominating Set Problem for Unit Disks Revisited. *International Journal of Computational Geometry and Applications*, 25(3):227, 2015.
- [4] M. Chellali and T. W. Haynes. A Note on the Total Domination Number of a Tree. *Journal of Combinatorial Mathematics and Combinatorial Computing*, 58:189, 2006.
- [5] V. Chvátal and C. McDiarmid. Small Transversals in Hypergraphs. *Combinatorica*, 12(1):19–26, 1992.
- [6] E. J. Cockayne, R. Dawes, and S. T. Hedetniemi. Total Domination in Graphs. *Networks*, 10(3):211–219, 1980.
- [7] E. J. Cockayne, M. A. Henning, and C. M. Mynhardt. Vertices Contained in all or in no Minimum Total Dominating Set of a Tree. *Discrete mathematics*, 260(1-3):37–44, 2003.
- [8] E. DeLaVina, Q. Liu, R. Pepper, B. Waller, and D. B. West. Some Conjectures of Graffiti. pc on Total Domination. 2007.
- [9] M. Dorfling, W. Goddard, and M. A. Henning. Domination in Planar Graphs with Small Diameter ii. 2006.
- [10] T. Haynes and M. Henning. Trees with Unique Minimum Total Dominating Sets. *Discussiones Mathematicae Graph Theory*, 22(2):233–246, 2002.
- [11] T. W. Haynes, S. Hedetniemi, and P. Slater. *Fundamentals of Domination in Graphs*. CRC press, 1998.
- [12] T. W. Haynes, S. T. Hedetniemi, and P. J. Slater. Domination in Graphs (Advanced Topics) Marcel Dekker Publications. New York, 1998.
- [13] M. A. Henning. A Survey of Selected Recent Results on Total Domination in Graphs. *Discrete Mathematics*, 309(1):32–63, 2009.
- [14] M. A. Henning and A. Yeo. *Total Domination in Graphs*. Springer, 2013.
- [15] D. S. Hochbaum and W. Maass. Approximation Schemes for Covering and Packing Problems in Image Processing and VLSI. *Journal of the ACM (JACM)*, 32(1):130–136, 1985.
- [16] J. Hopcroft and R. Tarjan. Efficient Planarity Testing. *Journal of the ACM (JACM)*, 21(4):549–568, 1974.
- [17] A. Itai, C. H. Papadimitriou, and J. L. Szwarcfiter. Hamilton Paths in Grid Graphs. *SIAM Journal on Computing*, 11(4):676–686, 1982.
- [18] D. Lichtenstein. Planar Formulae and Their Uses. *SIAM Journal on Computing*, 11(2):329–343, 1982.
- [19] R. Laskar, J. Pfaff, S. Hedetniemi, and S. Hedetniemi. On the Algorithmic Complexity of Total Domination. *SIAM Journal on Algebraic Discrete Methods*, 5(3):420–425, 1984.
- [20] J. Pfaff, R. Laskar, and S. Hedetniemi. Np-completeness of Total and Connected Domination and Irredundance for bipartite graphs. In *Tech. Rept. 428*. Clemson University Clemson, SC, 1983.
- [21] L. Sun. An Upper Bound for the Total Domination Number. *J. Beijing Inst. Tech*, 4(2):111–114, 1995.
- [22] S. Thomassé and A. Yeo. Total Domination of Graphs and Small Transversals of Hypergraphs. *Combinatorica*, 27(4):473–487, 2007.
- [23] Z. Tuza. Covering all Cliques of a Graph. *Discrete Mathematics*, 86(1-3):117–126, 1990.
- [24] T. Nieberg and J. Hurink. A PTAS for the minimum dominating set problem in unit disk graphs. *International Workshop on Approximation and Online Algorithms*, 296–306, 2005.
- [25] B. N. Clark, C. J. Colbourn and D. S. Johnson. Unit disk graphs. *Discrete mathematics*, 86(1-3):165–177, 1990.
- [26] M. V. Marathe, H. Breu, H. B. Hunt III, S. S. Ravi and D. J. Rosenkrantz. Simple heuristics for unit disk graphs. *Networks*, 25(2):59–68, 1995.
- [27] C. Ambühl, T. Erlebach, M. Mihalák and M. Nunkesser. Constant-factor approximation for minimum-weight (connected) dominating sets in unit disk graphs. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques*, 3–14, 2006.
- [28] T. Erlebach and M. Mihalák. A $(4 + \epsilon)$ -approximation for the minimum-weight dominating set problem in unit disk graphs. In *International Workshop on Approximation and Online Algorithms*, 135–146, 2009.

- [29] H. B. Hunt III, M. V. Marathe, V. Radhakrishnan, S. S. Ravi, D. J. Rosenkrantz, and R. E. Stearns. NC-Approximation schemes for NP- and PSPACE-hard problems for geometric graphs. *Journal of Algorithms*, 26(2):238–274, 1998.
- [30] H. Breu, and D. G. Kirkpatrick. Unit disk graph recognition is NP-hard. *Computational Geometry*, 9(1-2):3–24, 1998.
- [31] T. Erlebach, K. Jansen, and E. Seidel. Polynomial-time approximation schemes for geometric graphs. In *SODA*, 1: 671–679, 2001.
- [32] L. G. Valiant. Universality Considerations in VLSI Circuits. *IEEE Transactions on Computers*, 100(2):135–140, 1981.